Madeleine (Maddie) Wisinski

Syracuse, NY (Willing to Relocate) | maddie.wisinski@gmail.com | (469) 260 - 7881 | *www.wisinski.dev*

EDUCATION

Tulane University, New Orleans, Louisiana

- Majors: Engineering Physics (ABET) and Computer Science with a certificate in Computational Engineering

---

WORK EXPERIENCE

Novanta **July 2024 - Present**

Manufacturing Engineer

- Resolved critical factory issues related to equipment performance by identifying the source problems and implementing effective corrective actions, contributing to improved production output and reduced delays
- Diagnose faulty electronics using a multimeter and oscilloscope in the process of root cause analysis
- Refactored legacy code for an internal application, eliminating a persistent 2.5 minute delay
- Devised testing scripts for certain SKUs to reinforce longevity and reduce manual operations in production
- Designed an automatic testing fixture for handheld scanning device using Arduino C++ in PlatformIO

Tulane MakerSpace **June 2022 - May 2024**

Fabrication Technician

- Assessed and assisted on unique engineering problems from hundreds of students, faculty, and staff
- Repaired and maintained machines, such as the Bambu/Creality 3D printers and Epilog Laser Engravers
- Trained students on the correct and safe use of machines to support education and creativity

Northwestern Mutual **June 2023 - August 2023**

Software Engineer Intern

- Coordinated with teammates in an Agile environment on a large-scale frontend ReactJS application
- Increased testing coverage to over 80% through unit, functional, and regression tests
- Pinpointed issues with teammates while while refactoring old code to maintain stability
- Utilized Git Version Control and Kubernetes effectively in a complex code environment

Tulane University **January 2023 - May 2024**

Computing Concepts (MATLAB) Teaching Assistant

- Facilitated one-on-one learning with students on the basics of programming logic
- Managed assignment grades and proctoring exams as well as teaching when requested

---

PROJECTS

Handheld Scanner Automatic Testing Fixture **December 2024 - Present**

Hardware & Firmware

- Develop a fixture that executes commands, replaces a manual testing procedure, and saves $46,000/yr in labor
- Collaborate with 2 teammates to effectively design a cohesive, widely-applicable testing fixture
- Analyze and compare data sheets of components to match desired functions and ESP32 MCU capabilities
- Utilize dual core to operate an I2C color sensor and an analog microphone simultaneously
- Implement a CI/CD pipeline process, as well as unit/functional testing using Unity for PlatformIO

Mesh Network Measurement Nodes **August 2023 - May 2024**

Team Leader, Lead Product Designer, & Firmware Engineer

- Led a team of 5 to design, manufacture, and deploy a mesh network of 100 custom sensors to aid reforestation
- Engineered a reusable and testable C++ firmware codebase for managing power, measurements, and meshed radio communication outdoors
- Designed, iterated, and tested weatherproof sensor housings, emphasizing efficient manufacturing processes
- Awarded $4,000 in funding from Tulane University's Office of Academic Enrichment & Engineering Physics

---

SKILLS

Programming: C, C++, Java, C#, Python, Arduino, JavaScript, NodeJS, ReactJS, CSS, HTML, Julia, MATLAB, Simulink Controls Systems, MATLAB PDE Toolbox, Git Version Control, Azure Web Apps, and OpenGL in C++
Machinery: 3D Printers, Epilog CO_2 Laser Engraver, CNC Mill & Lathe, Waterjet, and Woodshop & Metalshop Tools
Electronic Tools: Multimeter, Oscilloscope
Programs: Autodesk Fusion 360 (4 years), SolidWorks, Adobe Illustrator, and Microsoft Suite
Languages: English (Fluent), French (Intermediate)